

DEBARSHI ROY

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[!\[\]\(919a2cb85b99741a73c0c31a427236a8_img.jpg\) LinkedIn](#) | [!\[\]\(c9cd5a1c35167a83f09a35036fe5dcbd_img.jpg\) GitHub](#) | [!\[\]\(ae1936640fabdea8c18f922ca69733fe_img.jpg\) Google Scholar](#) | [!\[\]\(e81307241bb070bc7c1be4e4328b2244_img.jpg\) ORCID](#) | [!\[\]\(5145ac5c495d0d3391897543e0ba7223_img.jpg\) Website](#)

Kolkata, West Bengal - 700030, India

ACADEMIC EXPERIENCE

- **Indraprastha Institute of Information Technology Delhi**  5th May, 2025 - 4th July, 2025
Research Intern in the Microbiome Informatics Lab of Computational Biology Department New Delhi, Delhi, India
 - I worked on a project titled '**Disease-Oriented Analysis of Human Microbiome Metadata: Toward LLM-Based Predictive Models**', which focuses on leveraging large language models (LLMs) to develop predictive frameworks based on disease-specific patterns in human microbiome metadata.
- **Indian Institute of Technology Guwahati**  2nd January, 2024 - 20th December, 2024
Former Research Assistant in the MARS Lab of Computer Science & Engineering Department Guwahati, Assam, India
 - Worked for Teaching Assistant of the course "**Introduction to Computing**" (CS 101), Course Instructors: Prof. T. Venkatesh & Dr. John Jose, from January, 2024 - May, 2024.
 - Worked for Teaching Assistant of the course "**Digital Design & Computer Architecture**" (CS 322M), Course Instructor: Prof. Jatindra Kumar Deka, from July, 2024 - November, 2024.
 - [Experience Certificate](#)
- **Indian Institute of Science Education and Research Bhopal**  22nd July, 2023 - 31st December, 2023
Former JRF in the Electrical Engineering and Computer Science Department Bhopal, Madhya Pradesh, India
 - Worked for Teaching Assistant Position for the course **Discrete Mathematics (ECS 201)**, Course Instructor: Dr. Kundan Kandhway.

EDUCATION

- **Indraprastha Institute of Information Technology Delhi** July, 2025 - July, 2027
Master of Technology in Computational Biology New Delhi, Delhi, India
- **Maulana Abul Kalam Azad University of Technology** June, 2019 - June, 2023
Bachelor of Technology in Computer Science & Engineering Kolkata, West Bengal, India
 - CGPA: 8.80/10.00
 - [Marksheets & Degree Certificate](#)
- **Chakdaha Ramlal Academy** June, 2018
Higher Secondary Education (Class 12) Chakdaha, West Bengal, India
 - Marks (Percentage): 76%
 - [Marksheet & Certificate](#)
- **Gangnapur High (H.S.) School** May, 2016
Secondary Education (Class 10) Ranaghat, West Bengal, India
 - Marks (Percentage): 90.57%
 - [Marksheet & Certificate](#)

PROJECTS

- **Project A: METAQR: Image-Based Analysis in Microbiome** at IIIT Delhi
Tools: [Python, Flet, HTML/JS, Git]
 - Engineered a scalable clinical diagnostic tool that ultra-compresses massive metagenomic profiles (15 GB) into lightweight, byte-sized patient signatures.
 - Developed and deployed a serverless, cross-platform application (Web, iOS, Android) utilising dynamic QR codes for instantaneous point-of-care visualisation of gut ecology.
 - Processed over 70,000 patient samples to construct a highly optimised, low-memory clinical database, resulting in a formal patent filing for the system architecture.

• Project B: NEMESIS: Telemetry-Driven Distributed LLM Scheduler

at IIIT Delhi

Tools: [Python, Kubernetes, PyTorch DDP, Docker, Linux tc, Bash]

- Engineered a custom, multi-objective Kubernetes scheduler that intercepts default container placement to dynamically route LLM workloads, reducing execution time by 62% under severe CPU chaos.
- Implemented a topology-aware gang-scheduling algorithm that maps cluster latency matrices to mathematically protect PyTorch Distributed Data Parallel (DDP) synchronisation rings across data-centre racks.
- Integrated real-time NVIDIA NVML telemetry to bypass bare-metal GPU VRAM fragmentation, cutting compute times by over 40% and isolating network bottlenecks via deterministic latency sweeps.

• Project C: Machine Learning-based Age Prediction Model in gut microbial ageing-trajectories

at IIIT Delhi

Tools: [R, Python]

- Developing machine learning frameworks (Random Forest, XGBoost) to predict physiological age using gut microbiome data across multiple age cohorts (20-40, 41-80, 60-80).
- Analysing longitudinal aging trajectories to identify key microbial biomarkers, with Random Forest achieving the highest predictive accuracy for the 20-40 age demographic.
- Performing rigorous data filtering and correlation analysis to map microbial abundance changes to healthy aging patterns.

• Project D: Disease-Oriented Analysis of Human Microbiome Metadata: Toward LLM-Based Predictive Models at IIIT Delhi

Tools: [R, Python, Pandas, NumPy, Seaborn, Matplotlib, SciPy]

- Designing a predictive framework that harnesses Large Language Models (LLMs) to interpret unstructured patterns in human microbiome metadata associated with various diseases.
- Utilizing Pandas and SciPy for the statistical processing and dimensionality reduction of complex biological datasets to ensure high-quality model inputs.
- Creating visualization pipelines (Seaborn, Matplotlib) to uncover latent relationships between metadata features and disease states.

• Project E: A Survey Towards Secure, Adaptive, and Scalable Prefetching

in Association with IIT Guwahati

Tools: []

- Prepared a survey report on various Prefetching Mechanisms in the context of Cache Prefetching, Hybrid Software-Hardware Prefetching and Operating Systems Prefetching w.r.t I/O Devices.
- **Project Report**

PATENTS AND PUBLICATIONS

C=Conference, J=Journal, P=Patent, S=In Submission, T=Thesis

- [J.1] Debarshi Roy, Soumita Banerjee, Alisha Ansari, Sourav Goswami, Nidhi P. Raval, Kaustav Adhikary, and Tarini Shankar Ghosh (2026). **Artificial Intelligence (AI) in microbiome-directed biotherapeutics development**. *Progress in Molecular Biology and Translational Science*, Elsevier Inc. DOI: [10.1016/bs.pmbts.2026.01.005](https://doi.org/10.1016/bs.pmbts.2026.01.005)
- [J.2] Debarshi Roy, Nidhi P. Raval, Vindhya Regonda, Omprakash Shete, Kaustav Adhikary, Deepak Kandhal, Shivangi Verma, and Tarini Shankar Ghosh (2026). **Overlapping signatures of gut microbiome in aging and disease: Enrichment of medication-associated oral-microbiome-predominant microbes in the gut**. Submitted to *FEBS Letters* and currently under revision.
- [S.2] Debarshi Roy and Tarini Shankar Ghosh (2026). **Scalable Microbiome Network Inference: Mitigating Sparsity and Computational Bottlenecks in Random Effects Models**. *bioRxiv Preprint*. DOI: [10.64898/2026.03.27.714858](https://doi.org/10.64898/2026.03.27.714858).

ACHIEVEMENTS

• GATE CS/IT 2023 Qualified

Scorecard Validity: till 31st March, 2026

Organized by IIT Kanpur

- During my GATE CS/IT preparation, I studied all core Computer Science subjects according to the GATE CS/IT syllabus.
- **GATE CS/IT Scorecard of 2023**

SKILLS

- **Programming Languages:** Python, R, C, C++
- **Specialized Area:** Computational Metagenomics, Computational Genomics, Biological Data Analysis, BioStatistics, High-Performance Computing, Operating Systems, Computer Networks, Computer Organization & Architecture, DBMS
- **Other Tools & Technologies:** R & R Studio, Git & GitHub, gem5, Wireshark, Pandas, NumPy, Seaborn, Matplotlib, SciPy
- **Research Skills:** LaTeX, PowerPoint, Excell, Word

ACADEMIC SERVICE

- Volunteer, IEEE ANTS & AIORI Conference, IIT Guwahati (Dec 2024)
- Volunteer, IIT Guwahati Campus Immersion Program (June–July 2024)
- Volunteer, Pedagogy Training Workshop, IIT Guwahati (March 2024)

ADDITIONAL INFORMATION

Languages:

- English (Professional working proficiency)
- German (Professional working proficiency)
- Hindi (Limited working proficiency)
- Bengali (Native or bilingual proficiency)

Hobbies: Cricket, Chess, Swimming, Drama

REFERENCES

- **Dr. Tarini Shankar Ghosh**
Assistant Professor, Department of Computational Biology
Indraprastha Institute of Information Technology Delhi, New Delhi-110020, India
Email: tarini.ghosh@iiitd.ac.in
Relationship: MTech Supervisor in IIIT Delhi
- **Dr. Debraj Chatterjee**
Assistant Professor, Department of Computer Science & Engineering
Techno International, Newtown, Kolkata, West Bengal-700156, India
Email: debraj.chatterjee@tict.edu.in
Relationship: Guide & Mentor during Undergraduation
- **Dr. Tanmay Sanyal**
Assistant Professor, Department of Zoology
Krishnagar Government College, University of Kalyani, West Bengal-741101, India
Email: tanmaysanyal@gmail.com
Relationship: Guide & Mentor in various projects

ACKNOWLEDGEMENT

I affirm that the information provided in this Resume is true to the best of my knowledge and belief. I am prepared to provide necessary documentation or clarification if required.